

PAULT DOCUMENTATION

Architecture

Source: docs/ARCHITECTURE.md

Theme: Clean Technical

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Architecture

Overview

Pault is a macOS SwiftUI app backed by SwiftData. It exposes three primary surfaces that all read/write the same local store: the main window, a menu bar popover, and a global hotkey launcher.

Core surfaces

- **Main window** (ContentView): sidebar filters, search, and prompt editing with the inspector panel.
- **Menu bar popover** (MenuBarContentView): quick list with copy/paste actions and lightweight prompt creation.
- **Hotkey launcher** (HotkeyLauncherView): global search and actions opened with Cmd+Shift+P.
- **Preferences** (PreferencesView): login item, dock visibility, and launcher defaults.
- **Onboarding** (OnboardingView): first-run walkthrough shown from ContentView when `hasCompletedOnboarding` is false.

Data layer

- SwiftData ModelContainer is created in PaultApp with Prompt, Tag, and TemplateVariable.
- If the persistent store fails to load, the app falls back to an in-memory store.
- PromptService is the operation layer used by all three surfaces for CRUD, filtering, copy/paste, and tag mutations.

Key flows

- **Editing**: title/content edits are debounced in PromptDetailView before saving.
- **Template variables**: TemplateEngine scans prompt content for `{{variable}}`, syncs TemplateVariable rows, and resolves copied/pasted output with filled values.
- **Tags**: managed via the inspector panel with color selection and in-place creation.
- **Filtering**: sidebar supports all, recent, archived, and tag filters; search runs across title, content, and tags.
- **Launcher**: hotkey opens a floating panel; copy/paste actions write to the clipboard and optionally simulate paste.

Supporting components

- SidebarView: sidebar filters, search field, prompt list with context menus and empty states.
- InspectorView: tag management, favorite toggle, dates, and archive controls.
- TemplateVariablesView: editable fields and resolved preview for variables parsed from prompt content.
- TagPillView / TagPillsView: reusable tag pill components with TagColors enum for color mapping.
- CopyToast: floating toast notification confirming clipboard copy actions.
- AccessibilityHelper: centralized Accessibility permission checking and CGEvent paste simulation.
- FlowLayout: custom SwiftUI Layout for wrapping tag pills.

Platform-specific behavior

- Global hotkey registration uses Carbon (GlobalHotkeyManager).
- Clipboard access uses NSPasteboard.

- Paste simulation uses CGEvent via AccessibilityHelper and requires macOS Accessibility permission (prompted automatically on first use).

PaultCore note

PaultCore is a separate Swift package containing expanded models (workflows, variables, usage logs) that is not currently integrated with the macOS app target.

System overview diagram

Mermaid source and PNG export live in docs/diagrams/.

```
flowchart LR
    User((User))

    subgraph PaultApp["Pault macOS App"]
        Main["Main Window"]
        Menu["Menu Bar Popover"]
        Hotkey["Hotkey Launcher"]
        Prefs["Preferences"]
        Inspector["Inspector Panel"]
    end

    Store["SwiftData Store"]
    Clipboard["macOS Clipboard"]
    FrontApp["Frontmost App"]

    User --> Main
    User --> Menu
    User --> Hotkey

    Main <--> Store
    Menu <--> Store
    Hotkey <--> Store

    Main --> Clipboard
    Menu --> Clipboard
    Hotkey --> Clipboard

    Clipboard --> FrontApp
    Prefs --> PaultApp
    Inspector --> Main
```

PNG export target: docs/diagrams/exports/system-overview.png (generated via scripts/render_mermaid.py)